Adding and Subtracting in Scientific Notation

Question Bank - Additional 4.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, add or subtract. Write each answer in scientific notation.

1. $(5.3\times10^{7})+(2.4\times10^{7})$

7.7×10^{7}

2. $(9.6\times10^{-3})-(4.2\times10^{-3})$

5.4×10^{-3}

3. $(7.4\times10^{5})+(8.9\times10^{5})$

$16.3\times10^{5}=1.63\times10^{6}$

4. $(6.3\times10^{-7})-(5.9\times10^{-7})$

$0.4\times10^{-7}=4\times10^{-8}$

Point one place right, exponent down one.

5. $(3.8\times10^{9})+(6.1\times10^{8})$

4.41×10^{9}

$6.1\times10^{8}=0.61\times10^{9}$.

6. $(2.7\times10^{-2})-(9.0\times10^{-4})$

2.61×10^{-2}

$9.0\times10^{-4}=0.09\times10^{-2}$.

Real quantities.

7. Mars's diameter is about 6.79×10^{3} kilometers. Earth's is about 1.28×10^{4} kilometers.

a. What is the sum of the two diameters?

1.959×10^{4} ***km***

$6.79\times10^{3}=0.679\times10^{4}$.

b. How much larger is Earth's diameter than Mars's?

6.01×10^{3} ***km***

$1.28\times10^{4}=12.8\times10^{3}$ *, and* $12.8-6.79=6.01$.

8. A charity raised about 4.85×10^5 dollars in its first year and about 1.27×10^6 dollars in its second.

a. How much did it raise in the two years together?

1.755×10^6 ***dollars***

$4.85 \times 10^5 = 0.485 \times 10^6$.

b. How much more did it raise in the second year than in the first?

7.85×10^5 ***dollars***

$1.27 \times 10^6 = 12.7 \times 10^5$.

Working backwards.

9. Find the missing number in each equation.

a. $(5.2 \times 10^8) + (n \times 10^8) = 1.34 \times 10^9$

n = 8.2

$1.34 \times 10^9 = 13.4 \times 10^8$.

b. $(9.6 \times 10^{-4}) - (n \times 10^{-4}) = 3.8 \times 10^{-4}$

n = 5.8

c. $(6.5 \times 10^3) + (3.5 \times 10^3) =$ _____

1×10^4

The coefficients sum to exactly 10.

Reading someone else's work.

10. Malik writes $(3.2 \times 10^5) + (7.1 \times 10^7) = 1.03 \times 10^{13}$. Name the two mistakes he made, and give the correct sum.

He added the coefficients without first writing both numbers over the same power of ten, and he added the exponents. The correct sum is 7.132×10^7 ***.***

$3.2 \times 10^5 = 0.032 \times 10^7$ *, and* $7.1 + 0.032 = 7.132$ *. His* 10.3×10^{12} *was then repaired correctly to* 1.03×10^{13} *— the repair is the one thing he did right.*

11. Which is the value of $(6.8 \times 10^{-6}) + (5.4 \times 10^{-6})$, written in scientific notation?

- ● **A.** 1.22×10^{-5}
- ○ B. 12.2×10^{-12}
- ○ C. 12.2×10^{-6}
- ○ D. 1.22×10^{-12}

A

12. Select ALL of the expressions whose value is 2.4×10^6 .

- ☒ **A.** $(1.2 \times 10^6) + (1.2 \times 10^6)$
- ☐ B. $(1.8 \times 10^6) + (6 \times 10^6)$
- ☒ **C.** $(2.42 \times 10^6) - (2 \times 10^4)$
- ☒ **D.** $(1.8 \times 10^6) + (6 \times 10^5)$
- ☒ **E.** $(2.5 \times 10^6) - (1 \times 10^5)$
- ☐ F. $(2.5 \times 10^6) - (1 \times 10^6)$

A, C, D and E